

KINEMATICS WITH VIDEO EXAMPLES



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Kinematics

with Videos of Typical Problems

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TABLE OF CONTENTS

Licensing

1: One-Dimensional Kinematics

- 1.1: Prelude to One-Dimensional Kinematics
- 1.2: Displacement
- 1.3: Vectors, Scalars, and Coordinate Systems
- 1.4: Time, Velocity, and Speed
- 1.5: How Fast Do I Need To Run to Catch the Bus (Video Solution)
- 1.6: Acceleration
- 1.7: Overtaking on the Highway (Video Solution)
- 1.8: Motion Equations for Constant Acceleration in One Dimension
- 1.9: Romeo and Juliet Race To Each Other, When Shall They Meet (Video Solution))
- 1.10: Problem-Solving Basics for One-Dimensional Kinematics
- 1.11: Falling Objects
- 1.12: Graphical Analysis of One-Dimensional Motion
- 1.E: Kinematics (Exercises)

2: Two-Dimensional Kinematics

- 2.1: Prelude to Two-Dimensional Kinematics
- 2.2: Kinematics in Two Dimensions - An Introduction
- 2.3: Vector Addition and Subtraction- Graphical Methods
- 2.4: Vector Addition and Subtraction- Analytical Methods
- 2.5: Your Generic Ball Rolling Down Hill Problem (Video Solution)
- 2.6: Tom and Jerry Dive Into a Pool, Who Splashes First (Video Solution))
- 2.7: Projectile Motion
- 2.8: Snowball Fight Calculation (Video Solution)
- 2.9: Addition of Velocities
- 2.E: Two-Dimensional Kinematics (Exercises)

Index

Glossary

Detailed Licensing

Detailed Licensing

Licensing

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CHAPTER OVERVIEW

1: One-Dimensional Kinematics

Objects are in motion everywhere we look. Everything from a tennis game to a space-probe flyby of the planet Neptune involves motion. When you are resting, your heart moves blood through your veins. And even in inanimate objects, there is continuous motion in the vibrations of atoms and molecules. Questions about motion are interesting in and of themselves: *How long will it take for a space probe to get to Mars? Where will a football land if it is thrown at a certain angle?* But an understanding of motion is also key to understanding other concepts in physics. An understanding of acceleration, for example, is crucial to the study of force. Kinematics is the branch of classical mechanics which describes the motion of points, bodies, and systems of bodies without consideration of the masses of those objects, nor the forces that may have caused the motion.

[1.1: Prelude to One-Dimensional Kinematics](#)

[1.2: Displacement](#)

[1.3: Vectors, Scalars, and Coordinate Systems](#)

[1.4: Time, Velocity, and Speed](#)

[1.5: How Fast Do I Need To Run to Catch the Bus \(Video Solution\)](#)

[1.6: Acceleration](#)

[1.7: Overtaking on the Highway \(Video Solution\)](#)

[1.8: Motion Equations for Constant Acceleration in One Dimension](#)

[1.9: Romeo and Juliet Race To Each Other, When Shall They Meet \(Video Solution\)](#)

[1.10: Problem-Solving Basics for One-Dimensional Kinematics](#)

[1.11: Falling Objects](#)

[1.12: Graphical Analysis of One-Dimensional Motion](#)

[1.E: Kinematics \(Exercises\)](#)

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1.1: Prelude to One-Dimensional Kinematics

Objects are in motion everywhere we look. Everything from a tennis game to a space-probe flyby of the planet Neptune involves motion. When you are resting, your heart moves blood through your veins. And even in inanimate objects, there is continuous motion in the vibrations of atoms and molecules. Questions about motion are interesting in and of themselves: *How long will it take for a space probe to get to Mars? Where will a football land if it is thrown at a certain angle?* But an understanding of motion is also key to understanding other concepts in physics. An understanding of acceleration, for example, is crucial to the study of force.



Figure 1.1.1: The motion of an American kestrel through the air can be described by the bird's displacement, speed, velocity, and acceleration. When it flies in a straight line without any change in direction, its motion is said to be one dimensional. (credit: Vince Maidens, Wikimedia Commons)

Our formal study of physics begins with kinematics which is defined as the *study of motion without considering its causes*. The word “kinematics” comes from a Greek term meaning motion and is related to other English words such as “cinema” (movies) and “kinesiology” (the study of human motion). In one-dimensional kinematics and [Two-Dimensional Kinematics](#) we will study only the *motion* of a football, for example, without worrying about what forces cause or change its motion. Such considerations come in other chapters. In this chapter, we examine the simplest type of motion—namely, motion along a straight line, or one-dimensional motion. In [Two-Dimensional Kinematics](#), we apply concepts developed here to study motion along curved paths (two- and three-dimensional motion); for example, that of a car rounding a curve.

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1.2: Displacement

Learning Objectives

- Define position, displacement, distance, and distance traveled.
- Explain the relationship between position and displacement.
- Distinguish between displacement and distance traveled.
- Calculate displacement and distance given initial position, final position, and the path between the two.

These cyclists in Vietnam can be described by their position relative to buildings and a canal. Their motion can be described by their change in position, or displacement, in the frame of reference.



Figure 1.2.1: These cyclists in Vietnam can be described by their position relative to buildings and a canal. Their motion can be described by their change in position, or displacement, in the frame of reference. (credit: Suzan Black, Fotopedia)

Position

In order to describe the motion of an object, you must first be able to describe its position—where it is at any particular time. More precisely, you need to specify its position relative to a convenient reference frame. Earth is often used as a reference frame, and we often describe the position of an object as it relates to stationary objects in that reference frame. For example, a rocket launch would be described in terms of the position of the rocket with respect to the Earth as a whole, while a professor's position could be described in terms of where she is in relation to the nearby white board. In other cases, we use reference frames that are not stationary but are in motion relative to the Earth. To describe the position of a person in an airplane, for example, we use the airplane, not the Earth, as the reference frame.

Displacement

If an object moves relative to a reference frame (for example, if a professor moves to the right relative to a white board or a passenger moves toward the rear of an airplane), then the object's position changes. This change in position is known as displacement. The word “displacement” implies that an object has moved, or has been displaced.

Definition: Displacement

Displacement is the change in position of an object:

$$\Delta x = x_f - x_0,$$

where Δx is displacement, x_f is the final position, and x_0 is the initial position.

In this text the upper case Greek letter size Δ (delta) always means “change in” whatever quantity follows it; thus, size Δx means change in position. Always solve for displacement by subtracting initial position size x_0 from final position x_f .

Note that the SI unit for displacement is the meter (m) (see Section on [Physical Quantities and Units](#)), but sometimes kilometers, miles, feet, and other units of length are used. Keep in mind that when units other than the meter are used in a problem, you may need to convert them into meters to complete the calculation.

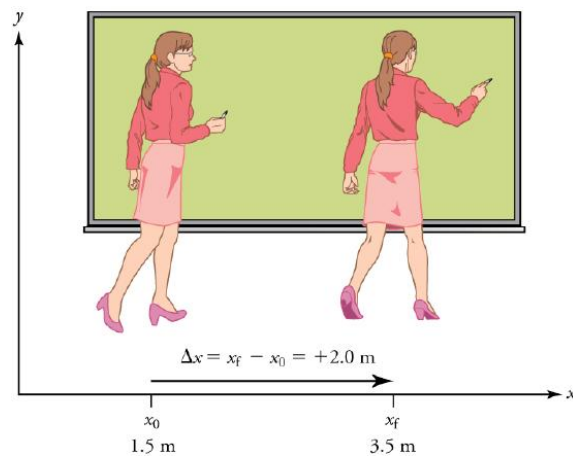


Figure 1.2.2: A professor paces left and right while lecturing. Her position relative to Earth is given by x . The +2 m displacement of the professor relative to Earth is represented by an arrow pointing to the right.

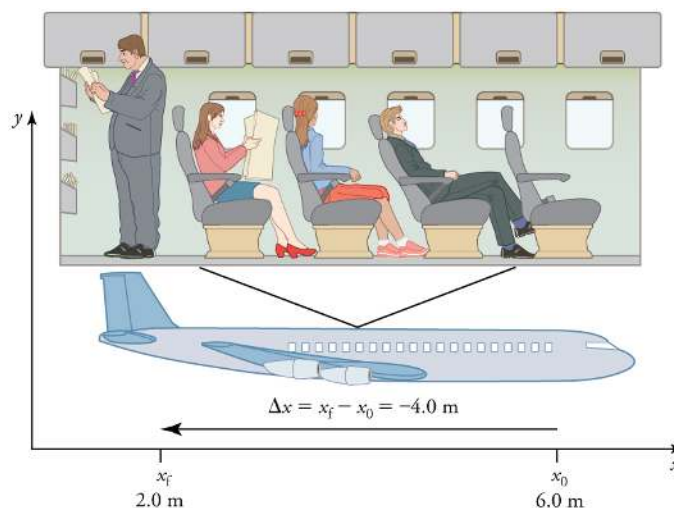


Figure 1.2.3: A passenger moves from his seat to the back of the plane. His location relative to the airplane is given by x . The -4 m displacement of the passenger relative to the plane is represented by an arrow toward the rear of the plane. Notice that the arrow representing his displacement is twice as long as the arrow representing the displacement of the professor (he moves twice as far)

Note that displacement has a direction as well as a magnitude. The professor's displacement in Figure 1.2.2 is 2.0 m to the right, and the airline passenger's displacement is 4.0 m toward the rear in Figure 1.2.3. In one-dimensional motion, direction can be specified with a plus or minus sign. When you begin a problem, you should select which direction is positive (usually that will be to the right or up, but you are free to select positive as being any direction). The professor's initial position is size $x_0 = 1.5 \text{ m}$ and her final position is $x_f = 3.5 \text{ m}$. Thus her displacement is

$$\begin{aligned}\Delta x &= x_f - x_0 \\ &= 3.5 \text{ m} - 1.5 \text{ m} \\ &= +2.0 \text{ m}.\end{aligned}$$

In this coordinate system, motion to the right is positive, whereas motion to the left is negative. Similarly, the airplane passenger's initial position is $x_0 = 6.0 \text{ m}$ and his final position is $x_f = 2.0 \text{ m}$, so his displacement is

$$\begin{aligned}\Delta x &= x_f - x_0 \\ &= 2.0 \text{ m} - 6.0 \text{ m} \\ &= -4.0 \text{ m}.\end{aligned}$$

His displacement is negative because his motion is toward the rear of the plane, or in the negative size $12\{x\}$ direction in our coordinate system.

Distance

Although displacement is described in terms of direction, distance is not. **Distance** is defined to be the *magnitude or size of displacement between two positions*. Note that the distance between two positions is not the same as the distance traveled between them. **Distance traveled** is the *total length of the path traveled between two positions*. Distance has no direction and, thus, no sign. For example, the distance the professor walks is 2.0 m. The distance the airplane passenger walks is 4.0 m.

Misconception Alert: Distance Traveled vs. Magnitude of Displacement

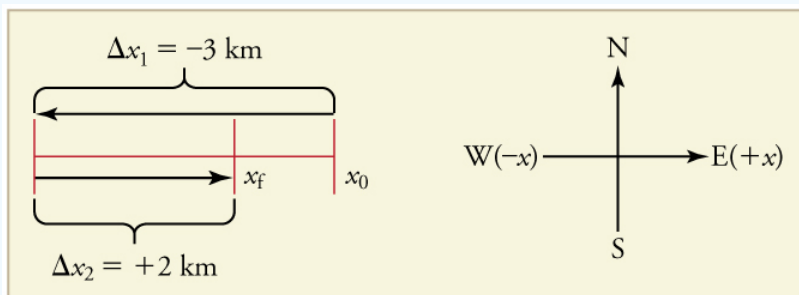
It is important to note that the distance traveled, however, can be greater than the magnitude of the displacement (by magnitude, we mean just the size of the displacement without regard to its direction; that is, just a number with a unit). For example, the professor could pace back and forth many times, perhaps walking a distance of 150 m during a lecture, yet still end up only 2.0 m to the right of her starting point. In this case her displacement would be +2.0 m, the magnitude of her displacement would be 2.0 m, but the distance she traveled would be 150 m. In kinematics we nearly always deal with displacement and magnitude of displacement, and almost never with distance traveled. One way to think about this is to assume you marked the start of the motion and the end of the motion. The displacement is simply the difference in the position of the two marks and is independent of the path taken in traveling between the two marks. The distance traveled, however, is the total length of the path taken between the two marks.

Exercise 1.2.1

A cyclist rides 3 km west and then turns around and rides 2 km east.

- What is her displacement?
- What distance does she ride?
- What is the magnitude of her displacement?

Answer



Answer a

The rider's displacement is $\Delta x = x_f - x_0 = -1 \text{ km}$. (The displacement is negative because we take east to be positive and west to be negative.)

Answer b

The distance traveled is $3 \text{ km} + 2 \text{ km} = 5 \text{ km}$.

Answer c

The magnitude of the displacement is 1 km .

Summary

- Kinematics is the study of motion without considering its causes. In this chapter, it is limited to motion along a straight line, called one-dimensional motion.
- Displacement is the change in position of an object.
- In symbols, displacement Δx is defined to be

$$\Delta x = x_f - x_0,$$

where x_0 is the initial position and x_f is the final position. In this text, the Greek letter Δ (delta) always means “change in” whatever quantity follows it. The SI unit for displacement is the meter (m). Displacement has a direction as well as a magnitude.

- When you start a problem, assign which direction will be positive.
- Distance is the magnitude of displacement between two positions.
- Distance traveled is the total length of the path traveled between two positions.

Glossary

kinematics

the study of motion without considering its causes

position

the location of an object at a particular time

displacement

the change in position of an object

distance

the magnitude of displacement between two positions

distance traveled

the total length of the path traveled between two positions

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